# PowerShell Byte Session SOP

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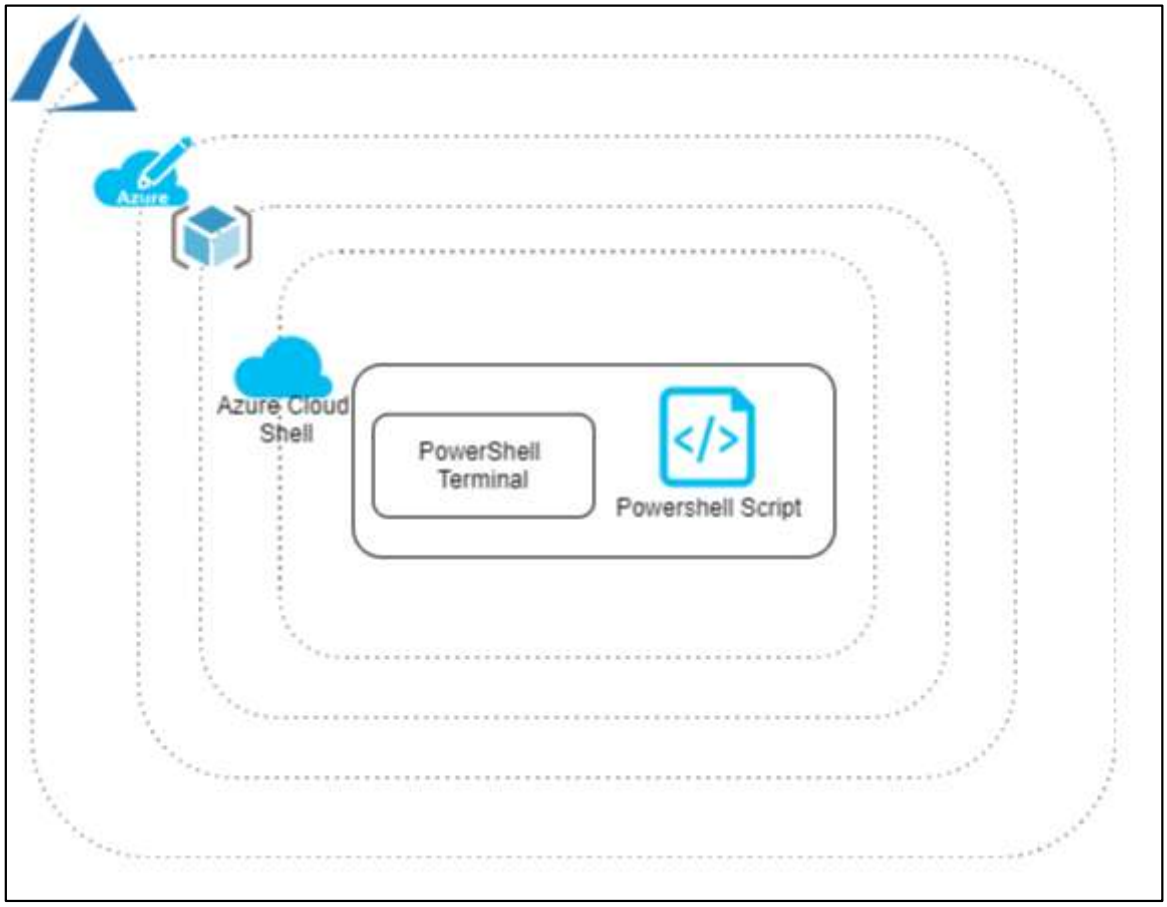
# 1. PowerShell Byte Session

## *1.1 Description*

PowerShell is a cross-platform task automation and configuration management framework, consisting of a command-line shell and scripting language. Unlike most shells, which accept and return text, PowerShell is built on top of the .NET Common Language Runtime (CLR), and accepts and returns .NET objects. Azure PowerShell is a set of cmdlets for managing Azure resources directly from the PowerShell command line.

## *1.2 Architecture Diagram*

The diagram below displays a visual representation of the application architecture:



### *1.3 Lab Steps*

Follow the steps outlined below to achieve the objective of this lab exercise:

1. Launch PowerShell in Azure Cloud Shell:
    - Go to the Azure Portal.
    - Click on the **Cloud Shell** button adjacent to Search bar.
    - Click on **Show Advanced Settings**.
    - Provide the following values:
        i. Cloud Shell region: Any (In case of any error, go as per the lab policy)
        ii. Resource Group: Select your Resource Group.
        iii. Storage Account Name: <any_unique_name>
        iv. FileShare: <any_unique_name>
        v. Click on **Create Storage**.
    - From the drop down, select **PowerShell** and **Confirm**.

2. Understanding PowerShell Basics:
   **PowerShell Cmdlet**: A cmdlet is also called as Command let which is a lightweight command used in the Window base PowerShell environment.
    - Following is some of the verbs use for you to learn PowerShell:
        - Get — To get something
        - Start — To run something
        - Out — To output something
        - Stop — To stop something that is running
        - Set — To define something
        - New — To create something
    - Important PowerShell Commands:
        - **Get-Command**: Brings up all the commands available for use in your current session.
        - **Set-ExecutionPolicy**: Enables you to control the level of security surrounding PowerShell scripts.
        - **Get-Service**: Helpful to know what services are installed on the system.
        - **Clear-History**: To clear the entries from your command history.
        - **Where-Object**: Enables you to take a dataset and pass it further down your pipeline for filtering.
        - **ConvertTo-SecureString**: Convert an encrypted standard string to a secure string or plain text to a secure string.
        - **Select-Object**: Selects the specified properties of a single object or group of objects.

3. Getting Help from PowerShell:
   - **Get-Help command**: Gives you an explanation about how to give a command and its parameter.
   - Example is shown in following Snippet: **Get-Help resourceGroup** will list down all command related to Resource Group.

```
PS /home/azureuser> Get-Help resourceGroup

Name                            Category  Module                 Synopsis
----                            --------  ------                 --------
Test-AzResourceGroup            Function  PSCloudShellUtility    …
Export-AzResourceGroup          Cmdlet    Az.Resources           Captures a resource group as a template and saves it to a file.
Get-AzResourceGroup             Cmdlet    Az.Resources           Gets resource groups.
Get-AzResourceGroupDeployment   Cmdlet    Az.Resources           Gets the deployments in a resource group.
Get-AzResourceGroupDeploymentOpe… Cmdlet  Az.Resources           Gets the resource group deployment operation
Get-AzResourceGroupDeploymentWha… Cmdlet  Az.Resources           Gets an ARM template What-If result for a deployment at resource group scope.
New-AzResourceGroup             Cmdlet    Az.Resources           Creates an Azure resource group.
New-AzResourceGroupDeployment   Cmdlet    Az.Resources           Adds an Azure deployment to a resource group.
Remove-AzResourceGroup          Cmdlet    Az.Resources           Removes a resource group.
Remove-AzResourceGroupDeployment Cmdlet   Az.Resources           Removes a resource group deployment and any associated operations.
Save-AzResourceGroupDeploymentTe… Cmdlet  Az.Resources           Saves a resource group deployment template to a file.
Set-AzResourceGroup             Cmdlet    Az.Resources           Modifies a resource group.
Stop-AzResourceGroupDeployment  Cmdlet    Az.Resources           Cancels a resource group deployment.
Test-AzResourceGroupDeployment  Cmdlet    Az.Resources           Validates a resource group deployment.
```

4. Working With PowerShell Pipelines:
   - Each command in a pipeline passes its output, item by item, to the next command.

```
PS /home/azureuser> Get-AzResourceGroup | where ResourceGroupName -Like
'CIS-PlatformServices-624062-RG'

ResourceGroupName : CIS-PlatformServices-624062-RG
Location          : centralus
ProvisioningState : Succeeded
Tags              :
                    Name        Value
                    ==========  ====================
                    Department  CIS-PlatformServices
                    CreatedFor  CCL

ResourceId        : /subscriptions/540b7602-d4c7-41d9-ae81-
9b1cfaa2f21d/resourceGroups/CIS-PlatformServices-624062-RG
```

   - In the above example, first command **Get-AzResourceGroup** lists down all the Resource Groups present in Azure portal.
   - The Pipeline symbol then passes this output to next command.
   - The Next command **where ResourceGroupName –Like** filters the Resource Group based on its name. So we get the required resource group from a list of resource groups with the help of Pipeline command.

5. Other PowerShell Commands Frequently Used in Azure:
   - **Connect-AzAccount**: To connect with Azure Account
   - **Get-AzVM**: To list VMs in a subscription
   - **New-AzVM -Name $myVM**: To create a Virtual Machine
   - **New-AzResourceGroup**: To create a new resource Group
   - **New-AzStorageAccount**: To create a new Storage Account

**Note**: Each of the command requires some parameters which you can easily find out with the PowerShell Help command.

6. Clean up resources:
   - When you are done with your lab exercise, delete all the resources created by you.

### *1.4 Troubleshooting*

| S. No | Problem | Solution |
|---|---|---|
| 1 | Not able to select desired Region | Go as per the Lab Policy, look for any error and suggested options to be followed. |

### *1.5 Supporting References*

Refer the below links for additional information:

1. https://docs.microsoft.com/en-us/powershell/azure/install-az-ps?view=azps-4.3.0
2. https://docs.microsoft.com/en-us/powershell/azure/get-started-azureps?view=azps-4.3.0
3. https://docs.microsoft.com/en-us/powershell/scripting/overview?view=powershell-7
4. https://github.com/Azure/azure-powershell